MATH 110 — Week 3

Notes Chapters 5–6; Homework 3

§1 Linear maps and their operations

The question. Which functions preserve linear combinations? The map $T(x, y) = (2x + y, x)$ does, since each output is a linear combination of the inputs.

Definition (Notes Definition 11; LADR 3.1)**.** Let U, V, W be vector spaces over the same field F. A map $T : V \to W$ is linear when it is additive, $T(v + w) = T(v) + T(w)$, and homogeneous, $T(av) = aT(v)$, for all $a \in F$ and $v, w \in V$.

Proposition (Notes Definition 11; LADR 3.10)

A linear map satisfies $T0 = 0$ and preserves every finite linear combination. Conversely, $T(av + w) = aTv + Tw$ for all a, v, w suffices for linearity.

Proof. 1. Additivity gives $T0 = T(0 + 0) = T0 + T0$; cancel $T0$ to obtain $T0 = 0$.

2. Homogeneity handles one term. Inductively apply additivity to a sum of the first k terms and the last term, obtaining $T(\sum_j a_j v_j) = \sum_j a_j T v_j$.
3. Conversely the combination test with $a = 1$ gives additivity, hence zero preservation as above; then $w = 0$ gives homogeneity.

□

Example (Notes Example 21)

For $n \geq 1$, differentiation $D : \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_{n-1}(\mathbb{R})$ is linear by the power rule and distributivity. We cite the elementary calculus facts that differentiation and integration are linear and that $(xf)' = f + xf'$.

1. On the complex vector space of continuous functions satisfying $f(t + 2\pi) = f(t)$, define $c_k(f) = \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-ikt}\, dt$ for each integer k. Here $e^{-ikt} = \cos(kt) - i \sin(kt)$. These are the Fourier coefficients; the integral of a complex function means the separate real and imaginary integrals.
2. Integral linearity gives $c_k(af + bg) = ac_k(f) + bc_k(g)$ for every k, so $f \mapsto (c_k(f))_{k \in \mathbb{Z}}$ is linear.

Example (LADR 3.3: linear maps on polynomial and coordinate spaces) 1. On $\mathcal{P}(\mathbb{R})$, $M_2 p = x^2 p$ is linear because $x^2(ap + q) = a(x^2 p) + x^2 q$.

2. For a fixed polynomial q, $C_q p = p \circ q$ is linear in p: $(ap + r)(q(x)) = ap(q(x)) + r(q(x))$. The polynomial q need not be linear; it is held fixed.
3. The map $J : \mathcal{P}(\mathbb{R}) \to \mathbb{R}$, $Jp = \int_0^1 p(t)\, dt$, is linear by integral linearity. For example $J(1 + 2x) = 1 + 2/2 = 2$.
4. Differentiation also defines a linear map on all of $\mathcal{P}(\mathbb{R})$, since each polynomial derivative is another polynomial.
5. The map $(x, y, z) \mapsto (2x - y + 3z, 7x + 5y - 6z)$ is linear: substituting $av + w$ into either coordinate and distributing gives a times that coordinate of Tv plus that of Tw.

Proposition (Notes Proposition 18; LADR 3.6)

$L(V,W)$, the set of linear maps $V \to W$, is a vector space under $(S+T)(v) = S(v)+T(v)$ and $(aT)(v) = aT(v)$.

Proof. 1. All maps $V \to W$ form a vector space pointwise: at each input the identities hold in W, just as in Week 1. The zero map is linear. For linear S, T, evaluate $(aS+T)(bv+w) = a(bS(v)+S(w)) + bT(v) + T(w)$.
2. Regrouping gives $b(aS+T)(v) + (aS+T)(w)$. Thus $aS+T$ is linear, proving the subspace condition.

□

Definition (Notes Definition 12; LADR 5.1)**.** An operator is a linear map $V \to V$; write $L(V) = L(V,V)$. The identity is $I_V(v) = v$, and the zero operator sends every vector to zero. Composition is $ST = S \circ T$, so T acts first. For an operator, $T^0 = I_V$ and $T^{k+1} = TT^k$.

Proposition (Notes Proposition 19; LADR 3.7–3.8)

If $T : U \to V$ and $S : V \to W$ are linear, then ST is linear. Composition is associative, has identities $TI_U = I_V T = T$, and distributes on both sides over sums of compatible maps.

Proof. 1. $ST(au+v) = S(aTu+Tv) = aSTu + STv$, proving linearity.
2. For compatible R, S, T, $((RS)T)u = R(S(Tu)) = (R(ST))u$. Also $(TI_U)u = Tu = (I_V T)u$.
3. $((S_1+S_2)T)u = S_1Tu + S_2Tu$ and $(S(T_1+T_2))u = ST_1u + ST_2u$ by linearity of S.

□

Example (Notes Examples 22–23) 1. On real sequences indexed by $0, 1, \ldots$, the shift $S(a_0, a_1, \ldots) = (a_1, a_2, \ldots)$ is linear coordinatewise. If $a_{n+2} = a_{n+1} + a_n$, its shifted sequence satisfies the same relation; hence on this recurrence subspace $S^2 - S - I = 0$ by comparison of every coordinate.
2. On $\mathbb{R}^2$, $P(x,y) = (0,y)$, $R_\theta(x,y) = (x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta)$, and $N(x,y) = (y,0)$ are linear by distributing each coordinate.
3. $R_{-\pi/2}P(x,y) = (y,0) = N(x,y)$ and $N^2(x,y) = N(y,0) = 0$. On the standard basis, $e_2 \mapsto e_1 \mapsto 0$.

$$e_2 \xrightarrow{N} e_1 \xrightarrow{N} 0$$

The second application of N sends both standard basis vectors to zero.

Example (Notes Example 24)

A smooth function is one with derivatives of every order. On smooth real functions define $Xf(x) = xf(x)$ and $Df = f'$. These are linear and again produce smooth functions by elementary calculus.

1. The product rule gives $(DX)f = (xf)' = f + xf'$, whereas $(XD)f = xf'$.
2. Therefore $DX - XD = I$. This is nonzero because $I(1) = 1$; thus two operators need not commute.

For the polynomial map $M_2 p = x^2 p$ above, the product rule similarly gives $(DM_2 - M_2 D)p = 2xp$ (LADR 3.9). Taking $p = 1$ gives $2x \neq 0$, so these maps do not commute.

Shortcuts

- Check $T(av+w) = aTv + Tw$ once; it gives both linearity conditions.
- To disprove linearity, first test $T(0) = 0$; every linear map satisfies it.

Traps

- $T(x) = x + 1$ is not linear since $T(0) = 1$. Passing the zero test alone is insufficient: $T(x) = x^2$ sends zero to zero but $T(2) = 4 \neq 2T(1)$.
- Composition order matters: for the maps above, $PN = 0$ while $NP = N \neq 0$.

§2 Constructing a map from specified values

The question. Can images of basis vectors be chosen freely? On $\mathbb{R}^2$, specifying $Te_1 = u$ and $Te_2 = w$ forces $T(x, y) = xu + yw$.

Definition (Notes Definition 13)**.** For a subspace $U \subseteq V$, the restriction $T|_U : U \to W$ has the same rule $T|_U(u) = T(u)$ and is linear because $au + v \in U$ and T is linear.

Proposition (Notes Proposition 20)

For a direct sum $U_1 \oplus \cdots \oplus U_m$ and linear maps $T_j : U_j \to W$, there is a unique linear map T on the sum whose restriction to each U_j is T_j.

Where the proof comes from

Want. A single linear map with the specified rule on each summand.

Know. Directness gives a unique decomposition $v = \sum_j u_j$, $u_j \in U_j$ (Week 1, Proposition 7).

Key step. Define $Tv = \sum_j T_j u_j$. Unique summands prevent two different outputs for the same input; addition and scaling act on those summands separately.

Proof. 1. Define $T(u_1 + \cdots + u_m) = \sum_j T_j(u_j)$. Unique components make this well-defined.
2. The components of $a\sum u_j + \sum v_j$ are $au_j + v_j$, so their images sum to $a\sum T_j u_j + \sum T_j v_j$. This proves linearity.
3. If only the jth component is nonzero, the rule agrees with T_j. Any linear extension must add those prescribed component images, proving uniqueness.

□

Proposition (Notes Proposition 21; LADR 3.4)

For a basis $v_1, \ldots, v_n$ of V and arbitrary $w_1, \ldots, w_n \in W$, there is a unique $T \in L(V, W)$ with $Tv_j = w_j$.

Proof. 1. A basis gives $V = Fv_1 \oplus \cdots \oplus Fv_n$: basis coefficients exist and are unique (Week 2). On Fv_j, prescribe $T_j(av_j) = aw_j$. Since $v_j \neq 0$, its coefficient a is unique; the rule is linear by adding/scaling coefficients.
2. Proposition 20 now gives the unique extension, explicitly $T(\sum_j a_j v_j) = \sum_j a_j w_j$. When $n = 0$, $V = \{0\}$ and its unique linear map sends zero to zero.

□

Proposition (Prescribing images on a spanning list)

Suppose $v_1, \ldots, v_r$ span V. A linear map with $Tv_j = w_j$ exists exactly when every relation $\sum_j a_j v_j = 0$ also satisfies $\sum_j a_j w_j = 0$. When it exists, it is unique.

Proof. 1. Necessity follows by applying T to a zero relation. For sufficiency, define $T(\sum_j a_j v_j) = \sum_j a_j w_j$. Subtracting two representations of the same input gives a zero relation, so the condition makes their proposed outputs equal.
2. Addition and scaling of representations show linearity. Every input has a representation because the list spans, and linearity forces the formula, proving uniqueness. A basis has only the zero relation, which explains why its images are unrestricted.

□

Prescribing images on a nonstandard basis

1. In $\mathbb{R}^2$, let $v_1 = (1, 1), v_2 = (1, -1)$; solving $(x, y) = av_1 + bv_2$ gives $a = (x + y)/2, b = (x - y)/2$. Thus the list is a basis.
2. Prescribe $Tv_1 = (1, 0)$ and $Tv_2 = (0, 2)$. Proposition 21 gives $T(x, y) = ((x + y)/2, x - y)$; substituting v_1, v_2 checks both values.

Shortcuts

- Two linear maps agreeing on a basis agree everywhere by Proposition 21.
- To define a map on a direct sum, define it independently on each summand; unique decomposition makes the rules compatible.

Traps

- Images of a dependent list cannot be freely prescribed: if $v_3 = v_1 + v_2$, linearity forces $Tv_3 = Tv_1 + Tv_2$.
- A non-direct sum requires compatibility: assigning different values to the same nonzero vector in an intersection cannot define a function.

§3 Kernel, range and rank–nullity

The question. For $P(x, y) = (0, y)$, which inputs map to zero, and which vectors occur as outputs? The equations give $y = 0$ for the first question and outputs $(0, y)$ for the second.

Definition (Notes Definition 14; LADR 3.11, 3.16)**.** The kernel (null space) is $\ker T = \{v \in V : Tv = 0\}$; the range (image) is $\operatorname{range} T = \{Tv : v \in V\} \subseteq W$.

Proposition (Notes Proposition 22; LADR 3.13, 3.18)

For $T \in L(V, W)$, $\ker T$ is a subspace of V and $\operatorname{range} T$ is a subspace of W.

Proof. 1. Zero belongs to both sets. If $u, v \in \ker T$, then $T(au + v) = aTu + Tv = 0$.

2. If $w_1 = Tv_1$ and $w_2 = Tv_2$, then $aw_1 + w_2 = T(av_1 + v_2)$ lies in the range. The subspace test proves both claims. □

Definition (Notes Definition 15)**.** The dimensions of $\ker T$ and $\operatorname{range} T$ are called the nullity and rank of T, respectively.

Example (Notes Example 25)

For $D \in L(\mathcal{P}_4(\mathbb{R}))$, $Dp = p'$.

1. If $p = a_0 + a_1x + \cdots + a_4x^4$, then $Dp = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3$. Thus $Dp = 0$ forces $a_1 = 2a_2 = 3a_3 = 4a_4 = 0$, so $a_1 = \cdots = a_4 = 0$ over $\mathbb{R}$, while a_0 is free. Hence $\ker D = \operatorname{span}(1)$.
2. The images of the monomial basis are $0, 1, 2x, 3x^2, 4x^3$, which span $\mathcal{P}_3(\mathbb{R})$. Every image has degree at most 3, so $\operatorname{range} D = \mathcal{P}_3(\mathbb{R})$, nullity 1, rank 4.

Example (LADR 3.12, 3.17, 3.20: kernels, ranges and codomains) 1. The zero map has kernel V and range $\{0\}$ by its rule.

2. For $\phi : \mathbb{C}^3 \to \mathbb{C}$, $\phi(z_1, z_2, z_3) = z_1 + 2z_2 + 3z_3$, the equation $\phi(z) = 0$ gives $\ker \phi = \{(-2s - 3t, s, t) : s, t \in \mathbb{C}\}$. Every $c \in \mathbb{C}$ is $\phi(c, 0, 0)$, so the range is $\mathbb{C}$.
3. For $M_2 p = x^2 p$, a zero output has every coefficient of p zero, so $\ker M_2 = \{0\}$. Its range consists of the polynomials with zero constant and linear coefficients: they are exactly the multiples of x^2.
4. The backward shift on F^∞, $(x_1, x_2, \ldots) \mapsto (x_2, x_3, \ldots)$, has kernel $\{(a, 0, 0, \ldots) : a \in F\}$. It is onto because $(0, y_1, y_2, \ldots)$ maps to any target $(y_1, y_2, \ldots)$.
5. For $T : \mathbb{R}^2 \to \mathbb{R}^3$, $T(x, y) = (2x, 5y, x + y)$, its range is $\{(u, v, w) : w = u/2 + v/5\}$. Necessity follows by substitution; conversely choose $x = u/2, y = v/5$ for any vector satisfying this equation.
6. Differentiation on all of $\mathcal{P}(\mathbb{R})$ is onto: a polynomial $q = \sum_{k=0}^{r} b_k x^k$ is the derivative of $\sum_{k=0}^{r} b_k x^{k+1}/(k+1)$. Its kernel consists of the constants.
7. On $\mathcal{P}_5(\mathbb{R})$, differentiation is not onto $\mathcal{P}_5(\mathbb{R})$ because x^5 cannot be a derivative there. With codomain $\mathcal{P}_4(\mathbb{R})$ the same rule is onto, by that antiderivative formula. Surjectivity depends on the codomain.

Proposition (Equal outputs and preimages; Notes Proposition 23; LADR 3.15, 3.19)

For a linear $T : V \to W$,

$$Tu = Tv \iff u - v \in \ker T.$$

Thus T is injective iff $\ker T = \{0\}$, and surjective iff $\operatorname{range} T = W$. If $Tv_0 = w$, all inputs with output w are exactly $\{v_0 + k : k \in \ker T\}$; if $w \notin \operatorname{range} T$, there are none.

Where the proof comes from

Want. Compare two inputs that give the same output.

Key step. Subtract their outputs: $Tu - Tv = T(u - v)$. The resulting zero condition is precisely what the kernel tests. Fixing one input v_0 therefore turns the search for every other input into adding an arbitrary kernel vector.

Proof. 1. $Tu = Tv$ iff $T(u - v) = 0$, proving the displayed equivalence. A zero kernel forces $u = v$; conversely injectivity applied to $Tv = T0$ forces $v = 0$.

2. With $Tv_0 = w$, the same equivalence says $Tv = w$ iff $v - v_0 \in \ker T$, giving the complete solution set. Surjectivity is exactly the assertion that every target is in the range.
3. Over $\mathbb{F}_2$, if V is finite-dimensional and w is attained, each such input corresponds to one kernel vector. A kernel basis gives $\dim \ker T$ independent coefficients, each 0 or 1, so there are $2^{\dim \ker T}$ inputs.

□

Proposition (Notes Proposition 24; LADR 3.21)

If V is finite-dimensional and $T : V \to W$ is linear, a basis $u_1, \ldots, u_m$ of $\ker T$ extends to a basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ of V such that $Tv_1, \ldots, Tv_n$ is a basis of $\operatorname{range} T$. Hence $\dim V = \dim \ker T + \dim \operatorname{range} T$.

Where the proof comes from

Want. A basis of the range whose length can be counted.

Know. Extend a kernel basis $u_1, \ldots, u_m$ to a domain basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ (Week 2, Proposition 11).

Key step. The Tu_i are zero, so only the Tv_j are needed to span the range. If $\sum_j a_j Tv_j = 0$, then $\sum_j a_j v_j$ lies in the kernel and equals a combination of the u_i. Moving terms to one side gives a relation in the full domain basis, so every $a_j = 0$.

Proof. 1. The kernel is a finite-dimensional subspace by Week 2 Proposition 14. Choose its basis and extend it by Week 2 Proposition 11.

2. If $\sum a_j T v_j = 0$, then $\sum a_j v_j \in \ker T$, so it equals $\sum b_i u_i$. Independence of the full domain basis forces all $a_j, b_i = 0$.
3. For any $w = Tv$, expand $v = \sum b_i u_i + \sum a_j v_j$; then $w = \sum a_j T v_j$. This proves spanning.
4. The dimensions are respectively $m+n, m, n$, giving the formula. No finite-dimensional hypothesis on W is needed.

□

Proposition (Rank–nullity consequences: Notes Proposition 25; LADR 3.22, 3.24, 3.26, 3.28, 3.65)

Let $\dim V = n$ and $\dim W = m$ be finite, and $T : V \to W$ linear.

(a) $\dim \operatorname{range} T \le \min(n, m)$ and $\dim \ker T \ge n - m$.

(b) If $n = m$, injectivity, surjectivity and bijectivity are equivalent. If $n > m$, no such T is injective; if $n < m$, none is surjective.

(c) A homogeneous linear system (all right-hand sides zero) with more unknowns than equations has a nonzero solution. With more equations than unknowns, some right-hand sides have no solution.

Proof. 1. Put $r = \dim \operatorname{range} T$. The range lies in W, so $r \le m$ (Week 2: an independent list in a subspace is independent in its ambient space). Rank–nullity gives $r = n - \dim \ker T \le n$ and $\dim \ker T = n - r \ge n - m$.

2. If $n = m$, zero kernel gives range dimension m, hence full range by the Week 2 full-dimension subspace criterion (a basis of that length already spans W); full range gives kernel dimension $n - m = 0$. If $n > m$, the lower bound on kernel dimension is positive. If $n < m$, $r \le n < m$ prevents full range. Apply Proposition 23 in each case.
3. The left sides of q linear equations in s unknowns define a linear map $F^s \to F^q$. If $s > q$, its kernel is nonzero; if $s < q$, its range is proper, so some targets are unattained.

□

Example (LADR 3.23: a dimension test before calculation)

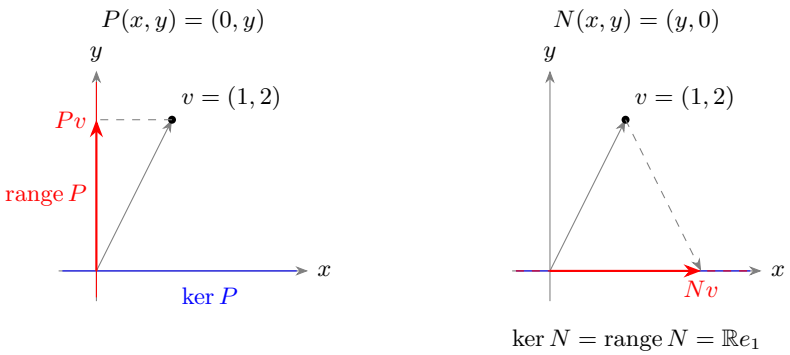
For $F = \mathbb{R}$ or $\mathbb{C}$, let

$$T(x_1, x_2, x_3, x_4) = (7x_1 + \pi x_2 + x_4,\ 97x_1 + 3x_2 + 2x_3,\ x_2 + 6x_3 + 7x_4).$$

Its kernel has dimension at least $4 - 3 = 1$. Thus T is not injective, without solving equations involving these coefficients. The dimension test establishes existence of a nonzero kernel vector; it does not give that vector's coordinates.

Compute kernel and range from a formula

1. For $P(x, y) = (0, y)$, $Pv = 0$ means $y = 0$, so $\ker P = \mathbb{R}e_1$; outputs are exactly $\mathbb{R}e_2$. These are complementary.
2. For $N(x, y) = (y, 0)$, the same equation $y = 0$ gives $\ker N = \mathbb{R}e_1$, but now $\operatorname{range} N = \mathbb{R}e_1$ too.
3. Both satisfy $2 = 1 + 1$, yet only the first gives a direct-sum decomposition.



For P, the kernel is the horizontal axis and the range is the vertical axis, giving a direct sum. For N, kernel and range are the same horizontal axis. Both maps have nullity 1 and rank 1; the dimension identity alone does not imply complementary subspaces.

For compatible linear maps $T : U \to V$ and $S : V \to W$,

$$ST = 0 \iff \operatorname{range} T \subseteq \ker S : \qquad S(Tu) = 0 \text{ for every } u \iff Tu \in \ker S \text{ for every } u.$$

For example, $N^2 = 0$ because $\operatorname{range} N = \mathbb{R}e_1 \subseteq \ker N$; $PN = 0$ for the same reason with $\ker P = \mathbb{R}e_1$.

Shortcuts

- To find a range, apply T to any spanning list: linearity says the images span the range.
- For equal finite dimensions, a zero kernel proves surjectivity without solving every output equation.

Traps

- Rank–nullity is a dimension identity, not $V = \ker T \oplus \operatorname{range} T$: the map N is a counterexample.
- Equal dimension is essential: $x \mapsto (x, 0)$ from $\mathbb{R}$ to $\mathbb{R}^2$ is injective but not onto. Finite dimension is essential too: the sequence shift is onto but has nonzero kernel.

§4 Coordinates and representation matrices

The question. For $N(x, y) = (y, 0)$, $Ne_1 = 0$ and $Ne_2 = e_1$ give the columns $(0, 0)^{\mathsf{T}}$ and $(1, 0)^{\mathsf{T}}$. How do such columns record a linear map in any pair of bases?

Definition (Notes Definition 16; LADR 3.29, 3.34, 3.36, 3.39)**.** An $m \times n$ matrix $A = (a_{ij})$ has m rows and n columns, with $a_{ij} \in F$ in row i, column j. The vector space $F^{m,n}$ uses entrywise addition and scaling. Its vector-space axioms hold entry by entry, by the field axioms. We write coordinate vectors as columns. The identity matrix I_n has ones on its diagonal and zeros elsewhere; its columns are $e_1, \ldots, e_n$.

Proposition (Notes Chapter 6; LADR 3.40)

The matrices E_{ij} with a single 1 at position (i, j) form a basis of $F^{m,n}$; hence its dimension is mn.

Proof. 1. Every A equals $\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} E_{ij}$, proving spanning.
2. The (i, j) entry of this sum is exactly a_{ij}; a zero sum forces every coefficient zero. There are mn basis matrices. □

Example (Notes Examples 26–27) 1. Entrywise operations give

$$\begin{pmatrix} 1+i & 3+2i & 4 \\ 5 & i & 2+i \end{pmatrix} + \begin{pmatrix} 4-2i & 5 & 1 \\ 3 & 2 & i \end{pmatrix} = \begin{pmatrix} 5-i & 8+2i & 5 \\ 8 & 2+i & 2+2i \end{pmatrix}.$$

2. Scaling gives $(1+i)\begin{pmatrix} 1+i & 3+2i & 4 \\ 5 & i & 2+i \end{pmatrix} = \begin{pmatrix} 2i & 1+5i & 4+4i \\ 5+5i & -1+i & 1+3i \end{pmatrix}$.
3. For $F^{2,2}$ the ordered basis is $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$.

Definition (Notes Definitions 18, 17; LADR 3.73, 3.31)**.** For an ordered basis $B = (v_1, \ldots, v_n)$, the coordinate column $[v]_B = (a_1, \ldots, a_n)^{\mathsf{T}}$ is defined by $v = \sum a_j v_j$. Here the superscript T converts a displayed row into a column. For a codomain basis $C = (w_1, \ldots, w_m)$, define $M_C^B(T)$ to have column j equal to $[Tv_j]_C$. Equivalently $Tv_j = \sum_i a_{ij} w_i$. The ordering of both bases matters.

Example (LADR 3.32: a rectangular representation matrix)

For $T : F^2 \to F^3$, $T(x, y) = (x + 3y, 2x + 5y, 7x + 9y)$,

$$Te_1 = \begin{pmatrix} 1 \\ 2 \\ 7 \end{pmatrix}, \qquad Te_2 = \begin{pmatrix} 3 \\ 5 \\ 9 \end{pmatrix}, \qquad M(T) = \begin{pmatrix} 1 & 3 \\ 2 & 5 \\ 7 & 9 \end{pmatrix}.$$

The two columns are the two basis images. Three output coordinates give three rows; two input coordinates give two columns.

Example (Notes Example 29) 1. For $p = 1 + 2x + 3x^2$, the coordinates in $(1, x, x^2)$ are $(1, 2, 3)^{\mathsf{T}}$.

2. The list $B = (1 + x, 1 + x^2, x + x^2)$ is independent: $a + b = 0$, $a + c = 0$, $b + c = 0$ force $2a = 0$, hence $a = b = c = 0$ over $\mathbb{R}$. Its length 3 gives a basis of $\mathcal{P}_2(\mathbb{R})$.
3. Since $p = 0(1 + x) + (1 + x^2) + 2(x + x^2)$, $[p]_B = (0, 1, 2)^{\mathsf{T}}$.

Build a representation matrix

1. Fix the ordered domain and codomain bases. Apply T to the first domain vector and express the result in the codomain basis; put those coefficients in column 1.
2. Repeat for each domain vector. For P, R_θ, N in the standard basis this gives

$$M(P) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad M(R_\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \quad M(N) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

3. The column count is the domain dimension; the row count is the codomain dimension.

Example (Notes Example 28)

Let $T : \mathcal{P}_3(\mathbb{C}) \to \mathbb{C}^4$ send p to $(p(1), p(i), p(-1), p(-i))$.

1. Evaluation is linear: $(ap + bq)(z) = ap(z) + bq(z)$ at each of the four points.
2. The four basis images are $T1 = (1, 1, 1, 1)$, $Tx = (1, i, -1, -i)$, $Tx^2 = (1, -1, 1, -1)$ and $Tx^3 = (1, -i, -1, i)$, using $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. The matrix below places these images in their respective columns.

$$A = M(T) = \begin{array}{c} \begin{matrix} T1 & Tx & Tx^2 & Tx^3 \end{matrix} \\ \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix} \end{array}$$

Each column is one basis image, evaluated in the order $1, i, -1, -i$. The column headings identify which input polynomial produces it.

Proposition (Notes Proposition 26; LADR 3.35, 3.38, 3.71–3.72)

For fixed bases of finite-dimensional V, W, $T \mapsto M_C^B(T)$ is a linear bijection $L(V, W) \to F^{m,n}$. Consequently $\dim L(V, W) = (\dim V)(\dim W)$.

Proof. 1. Every matrix gives unique vectors Tv_j by using its columns as C-coordinates. Proposition 21 gives exactly one map with those values, proving bijection.

2. The coordinates of $(S+T)v_j$ and aTv_j are the column sums and scalar multiples. Thus $M(S+T) = M(S)+M(T)$ and $M(aT) = aM(T)$.
3. The inverse images of the mn matrices E_{ij} form a basis: expand a matrix in them for spanning, and apply the injective correspondence to a zero relation for independence.

□

Shortcuts

- Apply a map only to basis vectors; Proposition 21 supplies all other values.
- Use polynomial coefficient comparison to find coordinates; a coefficient list is a coordinate vector only for the specified basis.

Traps

- Columns describe images, not rows: $N(e_2) = e_1$ places its 1 in row 1, column 2.
- The same polynomial has coordinates $(1, 2, 3)$ and $(0, 1, 2)$ above; changing a basis changes its coordinates.

§5 Matrix products and ranks

The question. Composing $t \mapsto (3t, 4t)$ with $(u, v) \mapsto u + 2v$ gives $t \mapsto 11t$. The corresponding row and column multiply as $(1\ 2)(3, 4)^{\mathsf{T}} = 3 + 8 = 11$. Why does this rule represent composition in general?

Definition (Notes Definitions 19–20; LADR 3.41, 3.44, 3.76)**.** For $A \in F^{m,n}$ and $x \in F^n$, set $(Ax)_i = \sum_{j=1}^{n} a_{ij}x_j$. For $B \in F^{n,r}$, define the $m \times r$ product by $(AB)_{ik} = \sum_{j=1}^{n} a_{ij}b_{jk}$. Both definitions depend only on the entries. The notation $A_{:j}$ means column j and $A_{i:}$ means row i. Distributivity of finite sums shows that $x \mapsto Ax$ is linear (the matrix-map part of Notes Example 21).

Example (LADR 3.42, 3.45: sizes and entries)

For $A = \begin{pmatrix} 8 & 4 & 5 \\ 1 & 9 & 7 \end{pmatrix}$, row 2 is $(1, 9, 7)$ and column 2 is $(4, 9)^{\mathsf{T}}$. In a product, each entry pairs one row with one column:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 6 & 5 & 4 & 3 \\ 2 & 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 10 & 7 & 4 & 1 \\ 26 & 19 & 12 & 5 \\ 42 & 31 & 20 & 9 \end{pmatrix}.$$

For instance, entry $(2, 1)$ is $3 \cdot 6 + 4 \cdot 2 = 26$. The sizes are $(3 \times 2)(2 \times 4) = 3 \times 4$; a matrix times a column is the case of one output column.

Proposition (Notes coordinate/product formulas; LADR 3.43, 3.46, 3.48, 3.50–3.51, 3.76)

For compatible bases, $[Tv]_C = M_C^B(T)[v]_B$ and $M_D^B(ST) = M_D^C(S)M_C^B(T)$. The kth product column is A times column k of B. Also $Ax = \sum_j x_j A_{:j}$. Each product column is a combination of columns of A; each product row is a combination of rows of B.

Where the proof comes from

Want. Find the coefficients of Tv and STv in the output basis.

Know. Column j records the coefficients of Tv_j.

Key step. Expand $v = \sum_j x_j v_j$, then $Tv = \sum_j x_j \sum_i a_{ij} w_i$. The coefficient of w_i is $\sum_j a_{ij}x_j$. Applying the same calculation to each Tv_k gives the matrix product formula.

Proof. 1. If $v = \sum_j x_j v_j$ and $Tv_j = \sum_i a_{ij} w_i$, then $Tv = \sum_i (\sum_j a_{ij}x_j) w_i$. This proves the coordinate identity and the column combination formula.

2. Apply this identity to $STv_k = S(Tv_k)$. Its coordinate column is $M(S)$ times column k of $M(T)$, yielding the product formula.

3. The entry formula gives $(AB)_{:k} = \sum_j b_{jk} A_{:j}$ and $(AB)_{i:} = \sum_j a_{ij} B_{j:}$. Here : means all entries of that column or row.

□

Proposition (Solvability and uniqueness from columns)

For $A \in F^{m,n}$, the system $Ax = b$ is solvable iff b lies in the span of the columns of A. Every target $b \in F^m$ is attainable iff the columns span F^m. Each attainable target has a unique input iff the columns are independent. For square A, spanning and independence are equivalent.

Proof. 1. The identity $Ax = \sum_j x_j A_{:j}$ gives the first two claims: solving is precisely expressing b as a column combination.

2. Column independence says $\sum_j x_j A_{:j} = 0$ forces $x = 0$. This is the zero-kernel criterion for uniqueness. In the square case, domain and codomain have equal finite dimension, so injectivity and surjectivity agree by rank–nullity.

□

Example (Notes Examples 30–31) 1. Row-times-column multiplication gives

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix} = \begin{pmatrix} 5+14+27 \\ 10+21+36 \end{pmatrix} = \begin{pmatrix} 46 \\ 67 \end{pmatrix}.$$

2. For the evaluation matrix A in Example 28, $A(1,2,1,0)^{\mathsf{T}} = (4, 2i, 0, -2i)^{\mathsf{T}}$. Directly evaluating $1 + 2x + x^2$ at $1, i, -1, -i$ gives the same four values.

Example (LADR 3.49: multiplication as a column combination)

The first output entry is $5 \cdot 1 + 1 \cdot 2 = 7$, the second $5 \cdot 3 + 1 \cdot 4 = 19$, and the third $5 \cdot 5 + 1 \cdot 6 = 31$:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 5 \\ 1 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} = \begin{pmatrix} 7 \\ 19 \\ 31 \end{pmatrix}.$$

The entries of the input column give the coefficients on the matrix's columns.

Example (Notes Example 32; LADR 3.33)

Let $T : \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ be differentiation and $S : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ send p to $\int_0^x p(t)\,dt$.

1. The power rule gives $M(T) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$ and $M(S) = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{pmatrix}$.
2. In $M(S)M(T)$ the nonzero diagonal entries are $1 \cdot 1$, $(1/2) \cdot 2$ and $(1/3) \cdot 3$, all equal to 1; the constant column is zero. Thus $M(ST) = M(S)M(T) = \operatorname{diag}(0, 1, 1, 1)$, where diag means the displayed diagonal entries and zeros elsewhere: $STp = p - p(0)$: for $p = a_0 + a_1 x + a_2 x^2 + a_3 x^3$, differentiating and taking the antiderivative without a constant returns $a_1 x + a_2 x^2 + a_3 x^3$.
3. Reversing the order gives $M(TS) = I_3$ and $TSp = p$. The two products even have different sizes.

Vectors representing data: Notes Chapter 6 A vector's entries can record features of an object, such as a token in a language model. For fixed W, the update is still the linear map $x \mapsto Wx$. For instance, $W = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}$ sends the feature vector $(2, 3, 1)^{\mathsf{T}}$ to $(2+1, 3-1)^{\mathsf{T}} = (3, 2)^{\mathsf{T}}$. Each output coordinate is a specified linear combination of the input coordinates.

Proposition (Notes Proposition 27; LADR 3.8, 3.43)

Matrix products are associative and distribute on both sides: $(AB)C = A(BC)$, $A(B + C) = AB + AC$, $(A + B)C = AC + BC$, whenever the sizes permit the operations. The identity matrix I_n has ones on its diagonal and zeros elsewhere; $AI_n = A$ and $I_m A = A$ for $A \in F^{m,n}$.

Proof. 1. The two associations send every x to the same vector $A(B(Cx))$. Likewise, $A((B+C)x) = ABx + ACx$ and $(A+B)(Cx) = ACx + BCx$, using linearity of the coordinate maps. Equality on every standard basis vector gives equality of their matrix columns.

2. $I_n x = x$ because its columns are e_j, so the same calculation gives $AI_n = A$ and $I_m A = A$.

□

Definition (LADR 3.52, 3.54, 3.58)**.** The column rank of A is the dimension of the span of its columns; its row rank is the dimension of the span of its rows (viewed as vectors). The transpose A^{T} exchanges rows and columns: $(A^{\mathsf{T}})_{ji} = a_{ij}$. We prove the two ranks agree and call their common value the rank of A.

Example (LADR 3.53, 3.55: rank and transpose) 1. Over $\mathbb{R}$ or $\mathbb{C}$, $A = \begin{pmatrix} 4 & 7 & 1 & 8 \\ 3 & 5 & 2 & 9 \end{pmatrix}$ has column rank 2: its first two columns are independent, since $4a + 7b = 3a + 5b = 0$ gives $b = 0$ by three times the first minus four times the second, then $a = 0$. For a relation between the rows, the first two entries give $4a + 3b = 7a + 5b = 0$; five times the first minus three times the second gives $-a = 0$, then $b = 0$. Thus row rank is also 2.

2. Transposition interchanges rows and columns: $\begin{pmatrix} 5 & -7 \\ 3 & 8 \\ -4 & 2 \end{pmatrix}^{\mathsf{T}} = \begin{pmatrix} 5 & 3 & -4 \\ -7 & 8 & 2 \end{pmatrix}$.

The column rank of an $m \times n$ matrix is at most n because it has n spanning columns, and at most m because those columns lie in F^m (Week 2, Proposition 14). The analogous row bound is $\min(m, n)$.

Proposition (LADR 3.54, 3C Exercises 14–15: transpose algebra)
For compatible matrices, $(A+B)^{\mathsf{T}} = A^{\mathsf{T}} + B^{\mathsf{T}}$, $(aA)^{\mathsf{T}} = aA^{\mathsf{T}}$ and $(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$.

Proof. 1. The (i, j) entries in the sum identity are both $a_{ji} + b_{ji}$; those in the scalar identity are both aa_{ji} (the first a is the scalar).

2. The (i, j) entry of $(AB)^{\mathsf{T}}$ is $\sum_k a_{jk} b_{ki} = \sum_k (B^{\mathsf{T}})_{ik} (A^{\mathsf{T}})_{kj}$, the corresponding entry of $B^{\mathsf{T}} A^{\mathsf{T}}$.

□

Proposition (LADR 3.56–3.57, 3.78)
If $A \in F^{m,n}$ has positive column rank c, then $A = CR$ where the c columns of $C \in F^{m,c}$ form a basis of the column span, and $R \in F^{c,n}$ contains the coefficients of the columns of A in that basis. Row rank equals column rank. Also $\operatorname{rank} M(T) = \dim \operatorname{range} T$.

Where the proof comes from

Want. Compare row rank with column rank c.

Know. Every column expands in a chosen column basis.

Key step. Put that basis in C and the expansion coefficients in R, giving $A = CR$. Every row of A is then a combination of the c rows of R, so row rank is at most c. Apply the inequality to A^{T} to obtain the reverse bound.

Proof. 1. Choose independent columns spanning all columns of A and put them in C. Put their expansion coefficients in R. The column formula for products gives $A = CR$.

2. Every row of $A = CR$ is a combination of the c rows of R. Thus row rank $\le c =$ column rank. For the zero matrix both ranks are zero.

3. Apply the inequality to A^{T}. Its rows are the columns of A and conversely, so column rank of $A \le$ row rank of A. Equality follows.

4. The coordinate map $w \mapsto [w]_C$ is linear and bijective by unique basis coefficients. The range is spanned by Tv_j; their coordinate columns are the columns of $M(T)$. This bijection preserves spanning and independence, so their dimensions agree.

□

Find a column–row factorization

1. Let $A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 1 & 1 & 5 \end{pmatrix}$. Its first two columns are independent by their first two entries; column 3 is twice column 1 plus three times column 2.
2. Therefore $C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $R = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{pmatrix}$ give $A = CR$, rank 2. The row relation is row 3 = row 1 + row 2.
3. Solving $Ax = 0$ gives $x_1 = -2x_3, x_2 = -3x_3$, so the kernel of the matrix map is $\operatorname{span}((-2, -3, 1))$, agreeing with $3 = 1 + 2$.

Shortcuts

- For AB, check inner dimensions first; the result has the outer dimensions.
- Use a product column $A(B_{:j})$ when only that column is needed.
- A column basis computes the rank and simultaneously gives a basis of the coordinate range.

Traps

- $AB = 0$ does not imply either factor is zero: $N^2 = 0$ but $N \neq 0$.
- Matrix multiplication is not entrywise: $N^2 = 0$, whereas squaring each entry of N returns N.
- Row rank equals column rank, but the row and column spaces usually live in different spaces: a 2×3 matrix has rows in F^3 and columns in F^2.

§6 Permutation and stochastic matrices

The question. If the present state is certainly A, and the next state is A or B with probability $1/2$ each, the next distribution on (A, B, C) is $(1/2, 1/2, 0)^{\mathsf{T}}$. How does a matrix record and combine these conditional distributions?

Definition (Notes Chapter 6)**.** A permutation σ of $\{1, \ldots, n\}$ is a bijection of this set. Its permutation matrix is defined by $S_\sigma e_j = e_{\sigma(j)}$: every row and column contains exactly one 1, all other entries being zero. A probability vector $p \in \mathbb{R}^n$ has nonnegative entries summing to 1. A column-stochastic matrix has nonnegative entries and each column sums to 1.

Proposition (Notes Chapter 6: repeated transitions)

$S_\sigma^k = S_{\sigma^k}$ for all integers $k \geq 0$. A real matrix P sends every probability vector to a probability vector iff it is column-stochastic. Products of compatible column-stochastic matrices are column-stochastic. A time-homogeneous finite Markov chain uses the same transition probabilities at every step and has next-state probabilities depending only on its present state; its distributions satisfy $p_k = P^k p_0$.

Where the proof comes from

Want. Compute repeated transitions and check total probability remains one.

Know. Column j records the output from the certain state e_j.

Key step. A permutation sends e_j successively to $e_{\sigma(j)}, e_{\sigma^2(j)}, \ldots$. For a stochastic matrix, sum $(Pp)_i$ over i and use $\sum_i P_{ij} = 1$ to obtain $\sum_j p_j = 1$.

Proof. 1. On e_j, $S_\sigma^k e_j = e_{\sigma^k(j)}$ by induction, proving the matrix identity by basis determination.

2. If p is a probability vector, $(Pp)_i = \sum_j P_{ij} p_j \geq 0$, and $\sum_i (Pp)_i = \sum_j (\sum_i P_{ij}) p_j = \sum_j p_j = 1$.
3. Conversely, each e_j is a probability vector. If P preserves all probability vectors, its column Pe_j is nonnegative and sums to 1, proving stochasticity. A product preserves probability vectors by applying its two factors successively, so the same criterion proves product closure.
4. Here P_{ij} is the probability of moving to i given present state j. Summing the mutually exclusive present states gives $p_{k+1} = Pp_k$; induction gives $p_k = P^k p_0$.

□

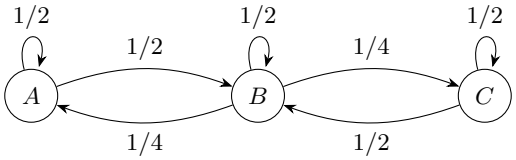
Example (Notes Example 33)

Take $P = \begin{pmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{pmatrix}$ and $p_0 = (1,0,0)^\mathsf{T}$.

1. Every column sums to 1 and is nonnegative. From state 1, the next state is 1 or 2 with probability $1/2$ each; column 2 gives probabilities $1/4, 1/2, 1/4$ from state 2.
2. The first column gives $p_1 = (1/2, 1/2, 0)^\mathsf{T}$. The next multiplication is

$$p_2 = \begin{pmatrix} (1/2)(1/2) + (1/4)(1/2) \\ (1/2)(1/2) + (1/2)(1/2) \\ (1/4)(1/2) \end{pmatrix} = \begin{pmatrix} 3/8 \\ 1/2 \\ 1/8 \end{pmatrix}.$$

   For p_3, the first entry is $(1/2)(3/8) + (1/4)(1/2) = 5/16$, the second is $(1/2)(3/8 + 1/2 + 1/8) = 1/2$, and the third is $(1/4)(1/2) + (1/2)(1/8) = 3/16$.
3. For $\pi = (1/4, 1/2, 1/4)^\mathsf{T}$, the first and third entries of $P\pi$ are both $(1/2)(1/4) + (1/4)(1/2) = 1/4$; the middle entry is $(1/2)(1/4 + 1/2 + 1/4) = 1/2$. Thus $P\pi = \pi$. Once this distribution is reached, it remains unchanged.



Column j of P lists the probabilities on arrows leaving state j, in the state order A, B, C. The outgoing probabilities at each state sum to 1; missing arrows have probability zero.

Switch between probability conventions

1. Transposing the coordinate equation $(Pp)_i = \sum_j P_{ij} p_j$ gives $(Pp)^\mathsf{T} = p^\mathsf{T} P^\mathsf{T}$.
2. Thus the row-vector convention uses the row-stochastic matrix P^T. Permutation matrices are column-stochastic and describe deterministic transitions $j \mapsto \sigma(j)$.

Shortcuts

- Check stochasticity column by column; preservation of total probability then follows without testing individual distributions.
- To verify an unchanged distribution, check $P\pi = \pi$ and that π is a probability vector.

Traps

- Row sums equal to 1 are insufficient for our column convention: $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right) e_1 = (1,1)^\mathsf{T}$ has total 2.
- An unchanged distribution need not be approached from other starting distributions: $I_n p = p$ for every p.

§7 Problem set

HW3 P1. Let $T : V \to W$ be linear with V finite-dimensional, and $U \subseteq W$ a subspace. (a) Prove $T^{-1}(U) = \{v : Tv \in U\}$ is a subspace. (b) Prove $\dim T^{-1}(U) = \dim \ker T + \dim(U \cap \operatorname{range} T)$. (c) For differentiation $D : \mathcal{P}_4(\mathbb{R}) \to \mathcal{P}_4(\mathbb{R})$, find the dimension of the polynomials satisfying $p'(x) = ax^2 + b$ for some $a, b \in \mathbb{R}$.
Hint: Apply rank–nullity to $T : T^{-1}(U) \to U \cap \operatorname{range} T$.

HW3 P2. (a) Find $T : \mathbb{R}^4 \to \mathbb{R}^4$ linear with $\ker T = \operatorname{range} T$. (b) Prove no such map exists on $\mathbb{R}^5$.
Hint: Use two coordinates as outputs; apply Proposition 24 to equal kernel and range dimensions.

HW3 P3. For $U = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = p'(1) = 0\}$, define $S : \mathcal{P}_4(\mathbb{R}) \to \mathbb{R}^2$ with kernel U. Use rank–nullity for $\dim U$ and compare with HW2.
Hint: Use the two constraints as outputs; test 1 and $x - 1$ for surjectivity.

HW3 P4. For linear $T : V \to W$, (a) prove a dependent list has a dependent image list, and (b) show an independent list need not have an independent image list.
Hint: Apply T to a nontrivial relation; use the zero map for (b).

HW3 P5. Let $T : \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_4(\mathbb{R})$ be $(Tp)(x) = x^2 p'(x) + p(x)$. (a) Find its matrix in the standard monomial bases. (b) Find kernel and range and their dimensions. (c) Explain directly why the range is the span of the representing columns, under monomial coordinates.
Hint: Apply T to $1, x, x^2, x^3$ and use Proposition 21.

HW3 P6. Over $\mathbb{F}_2$, solve $x + y + z + w = 1$, $x + z = 0$, listing every solution.
Hint: Solve for z, w in terms of x, y using addition modulo 2.

HW3 P7 (optional). Let A be an $n \times m$ matrix over $\mathbb{F}_2$ and suppose $Ax = b$ is consistent (has at least one solution). Express the number of solutions using m and $\operatorname{rank} A$.
Hint: Fix one solution x_0; every other solution differs by an element of $\ker A$. Count basis coefficients.

Exam practice (Practice Midterm 1, Question 2). Let V, W be finite-dimensional and $T : V \to W$ injective and linear. (a) For a basis $v_1, \dots, v_n$ of V, prove $Tv_1, \dots, Tv_n$ is independent. (b) Prove there exists a linear map $S : W \to V$ with $ST = I_V$. Such an S is called a left inverse of T.
Hint: Use injectivity for (a), then extend the image list to a basis and prescribe S there by Proposition 21.